ECON 0100 | HW E1 | Costs

Treat Homework as a tool to highlight what you don't yet understand. Grading is by submission, so there's no additional points for turning in perfect answers you don't fully understand. Use your areas of confusion to guide your study. Work in a group and ask others for help.

Q1 | Skyline Custom Brooms Production

The table shows how many workers it takes Skyline to make each number of brooms by hand. Workers are paid 240 coins a week. There are no fixed costs.

Workers	**0**	**1**	**2**	**3**	**4**
Brooms	0	8	14	18	20

a) What is the marginal product of the second worker?
b) What is the marginal product of the fourth worker?
c) As Skyline adds workers, does the marginal product rise, stay the same, or fall?
d) What is the variable cost of making 14 brooms a week?
e) What is the total cost of making 14 brooms a week?

Q2 | CloudSweep Brooms Production

The table shows how many workers it takes CloudSweep to make each number of brooms with standardized methods. Workers are paid 60 coins a week. Fixed costs are 1,000 coins a week.

Workers	**0**	**1**	**2**	**3**	**4**
Brooms	0	4	8	12	16

a) What is the marginal product of the second worker?
b) What is the marginal product of the fourth worker?
c) As CloudSweep adds workers, does the marginal product rise, stay the same, or fall?
d) What is the variable cost of making 8 brooms a week?
e) What is the total cost of making 8 brooms a week?

Q3 | Cost Structures

The Department of Magical Transportation is studying the broomstick manufacturing industry. Three major manufacturers dominate the market, each with distinct cost structures.

CloudSweep is an established manufacturer known for large facilities and expensive equipment (high fixed costs) and standardized production methods (constant marginal costs).

Skyline Custom manufactures personalized racing brooms from a basic workshop (no fixed costs) with each additional broom taking extra effort to make (increasing marginal costs).

SwiftTech is a new company using automated equipment it rents by the broom (no fixed costs) and standardized production steps (constant marginal costs).

a) Match curve in Row 1 of cost curve matrix to its manufacturer.

Column A: _____ Column B: _____ Column C: _____

b) Per Unit Subsidy

The Ministry gives firms 4 coins for each broom they make. Which graph shows SwiftTech's new cost curves?

Graph C1 Graph C2 Graph C3 Graph C4

c) Per Unit Tax

Which graph shows Skyline's cost curves after the Ministry imposes a 5 coin tax on each broom produced?

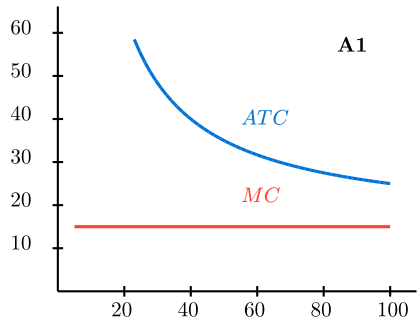
Graph B1 Graph B2 Graph B3 Graph B4

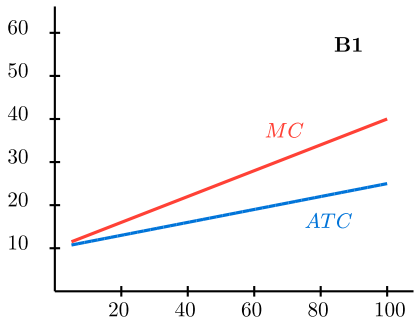
d) Lump Sum Tax

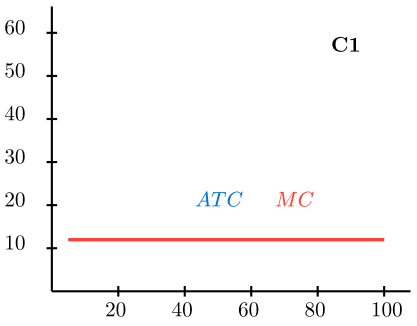
Which graph shows CloudSweep's cost curves after the Ministry imposes a 500 coin annual safety fee?

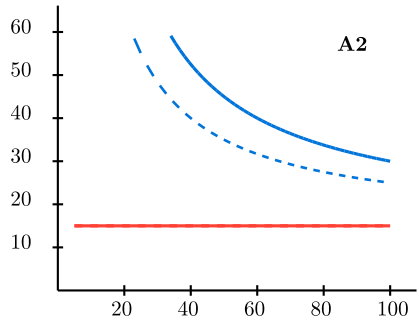
Graph A1 Graph A2 Graph A3 Graph A4

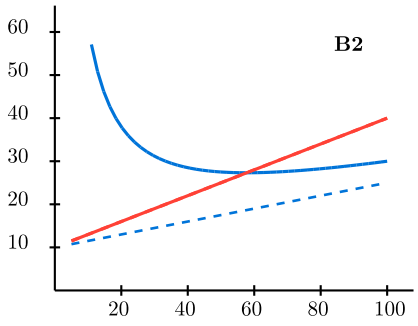
The graphs below show each manufacturer's ATC (blue) and MC (red). Row 1 is the base case. In Rows 2 to 4 the dashed curves repeat Row 1.

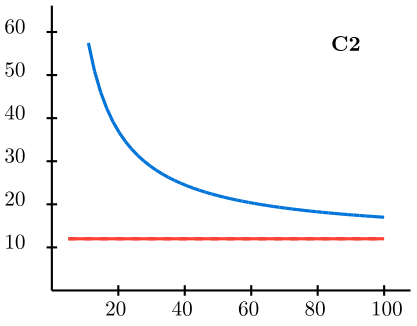

A1
ATC
MC
60
50
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60
80
100

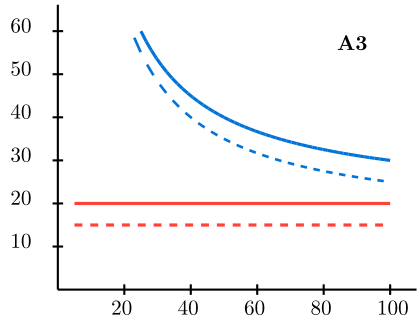

B1
MC
ATC
60
50
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60
80
100

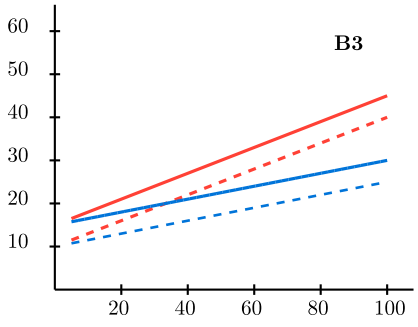

C1
ATC
MC
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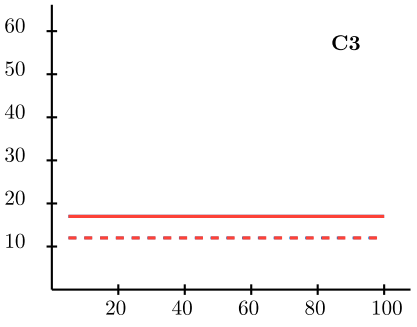

A2
60
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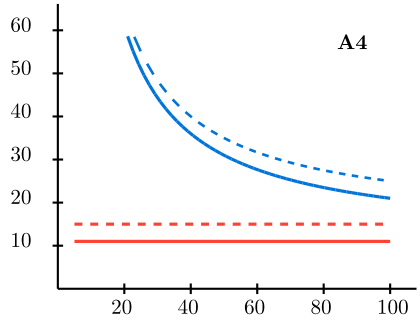

B2
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100

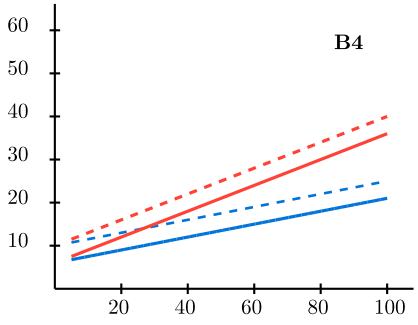

C2
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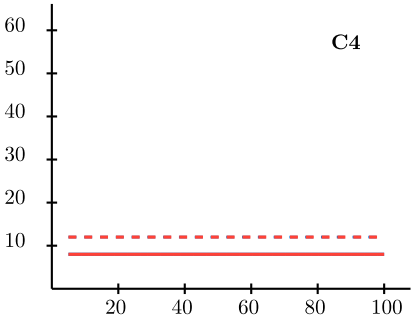

A3
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80
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B3
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10
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40
60
80
100


C3
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80
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A4
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80
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B4
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30
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10
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80
100


C4
60
50
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30
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10
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40
60
80
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